Suicide Rates in West Germany

[redacted]

CRJU 705

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Introduction

Suicides are an important criminological issue, and rates can vary across demographic groups and methods. In this project, I use data on suicides in West Germany to examine how suicide frequency differs by age, sex, and method of suicide. The data was collected from Heuer (1979) and provides detailed information that makes it possible to explore what might be connected to differences in suicide frequency. Identifying these trends is important because it helps reveal which groups may be more vulnerable and what kinds of cases come up more often. In general, the more we understand about when and how suicides take place, the better researchers, policymakers, and mental-health professionals can be at developing effective prevention strategies. To guide this analysis, the research question for this project is: *How do age, sex, and suicide method relate to suicide frequency in West Germany?* This project examines those trends systematically in an effort to provide insight into how social and demographic factors may bear on the frequency of suicide and to illustrate why continued research in this area is crucial to informing improved prevention efforts.

Methods

To examine how suicide frequency varies across demographic and situational factors in West Germany, I conducted a series of descriptive, visual, and statistical analyses using the Suicide_Germany_df dataset from the crimedatasets package. The variables used were freq (numeric variable representing the frequency of suicides), sex (factor indicating the sex of the individual), method (factor indicating the method of suicide out of 9 levels), and age (numeric variabe representing the age of the individual). First, I calculated the descriptive statistics by calculating means, frequencies, and distributions for age, sex, method, and suicide frequency.

This step helped establish a basic understanding of which demographic groups appeared most in the data and which methods were used more frequently. Then, I created several visualizations.

Figure 1

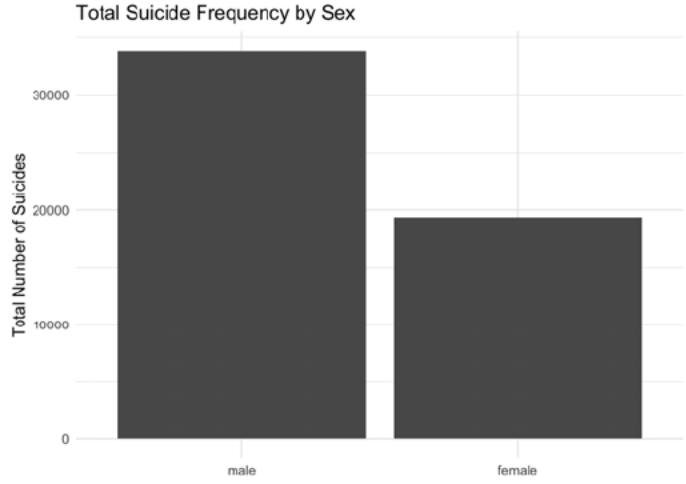


Figure 2

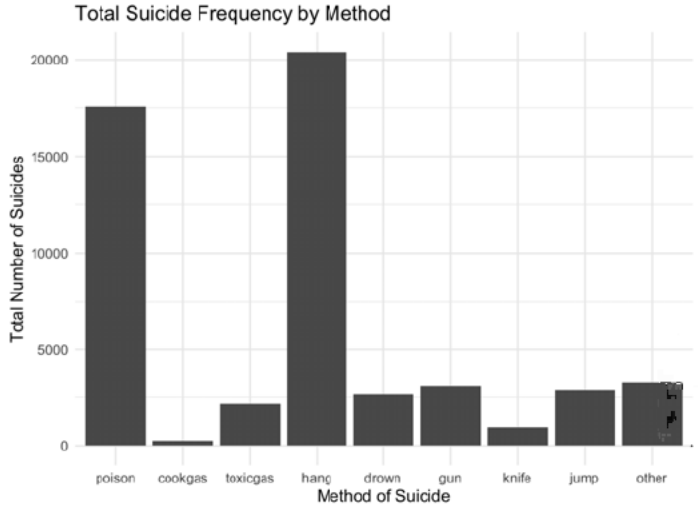
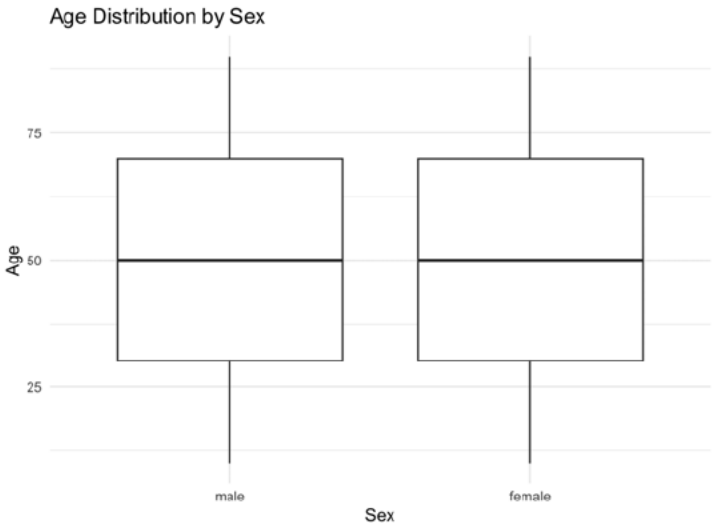


Figure 3



The first graph shows total suicide frequency by sex, showing the overall difference in suicide counts between males and females. The second graph shows the total number of suicides by method, which made it easy to compare how common each method was in the dataset. The third graph is a boxplot of age by sex, allowing me to see how age varies between men and women and whether one group tends to be older or younger in these cases. Lastly, I examined the relationship of the variables among one another through the use of measures of association. For the categorical variables, like sex and method of suicide, I constructed simple contingency tables to determine how often different combinations occurred. For continuous variables, like age and frequency of suicide, I computed correlations in order to get an idea of whether they had a tendency to rise or fall together. I created a linear regression to examine how age, sex, and suicide method relate to suicide frequency in the dataset, using the frequentist model. I expressed it as: Freq = Intercept + age + sex + method + Error.

Results

Figure 1 showed that males had far higher suicide counts than females. This shows that there is clear gender difference in suicide frequency within this data set. Figure 2 showed the number of suicides per method. Hanging was the most frequent with poison coming at a close second. But, the other 7 methods were substantially lower. Figure 3 showed a boxplot of age by sex, and the age patterns for men and women looked pretty similar overall. However, the range of ages among males was a bit wider, suggesting slightly more variation in men's ages in this sample.

The linear regression model shows that the reference group is poisoning. Age had an extremely small effect on suicide frequency (-0.7383) and was not strong enough to show a

statistically significant pattern. The coefficient for sex shows that females had about 94 fewer suicides than males on average, showing a very significant effect. his supports the pattern seen in the descriptive statistics, where males had much higher suicide frequencies. The regression results showed that almost every suicide method occurred far less frequently than the reference method. They all had very large negative coefficients, besides hanging. Hanging was the only positive coefficient, meaning that it was slightly more common than poisoning. Overall, the regression results confirm the descriptive patterns in as much as sex and method are significant predictors of the frequency of suicide, but age plays a much smaller role.

```
lm(formula = Freq ~ age + sex + method, data = suicide)

Residuals:
    Min      1Q  Median      3Q     Max
-589.39  -55.29  -11.65   54.50  730.74

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    600.7726    39.7032  15.132  < 2e-16 ***
age             -0.7383     0.4306  -1.714   0.0875 .
sexfemale      -94.4837    21.0970  -4.479 1.08e-05 ***
methodcookgas -509.1765    44.7536 -11.377  < 2e-16 ***
methodtoxicgas -453.2647    44.7536 -10.128  < 2e-16 ***
methodhang      82.7059    44.7536   1.848   0.0656 .
methoddrown   -438.7059    44.7536  -9.803  < 2e-16 ***
methodgun     -424.9118    44.7536  -9.494  < 2e-16 ***
methodknife   -489.0588    44.7536 -10.928  < 2e-16 ***
methodjump    -432.9412    44.7536  -9.674  < 2e-16 ***
methodother   -420.0294    44.7536  -9.385  < 2e-16 ***
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 184.5 on 295 degrees of freedom
Multiple R-squared:  0.5829,    Adjusted R-squared:  0.5688
F-statistic: 41.23 on 10 and 295 DF,  p-value: < 2.2e-16
```

Discussion

In conclusion, these findings helped me create a better understanding of the suicide rates in West Germany. It helped make sense of what factors are most strongly related to suicide frequency. The pattern between all the statistics show that sex and suicide method play a major role in explaining how often suicides occurred in this dataset. Males had far higher suicides than females. The method of suicide also mattered significantly, as poisoning and hanging were far

more common than the other types. This project showed me that studying this topic is extremely important, as suicide is prominent in todays world. Ultimately, these patterns show that both who a person is and the circumstances of the act play a role in suicide trends